# Architectural Skills

*CSSE6400*

Richard Thomas

May 18, 2026

*Quote*

Architecture is the stuff you can't Google.

– Mark Richards [Richards and Ford, 2020]

*Quote*

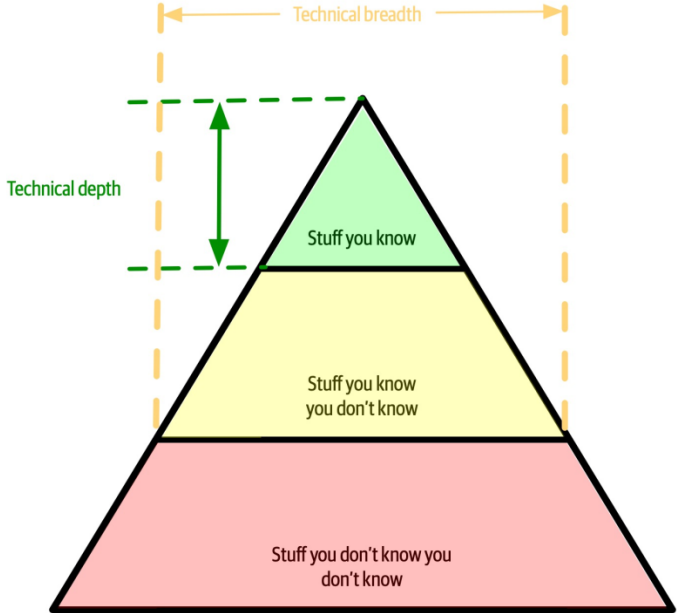
There are no right or wrong answers in architecture—only trade-offs.

– Neal Ford [Richards and Ford, 2020]

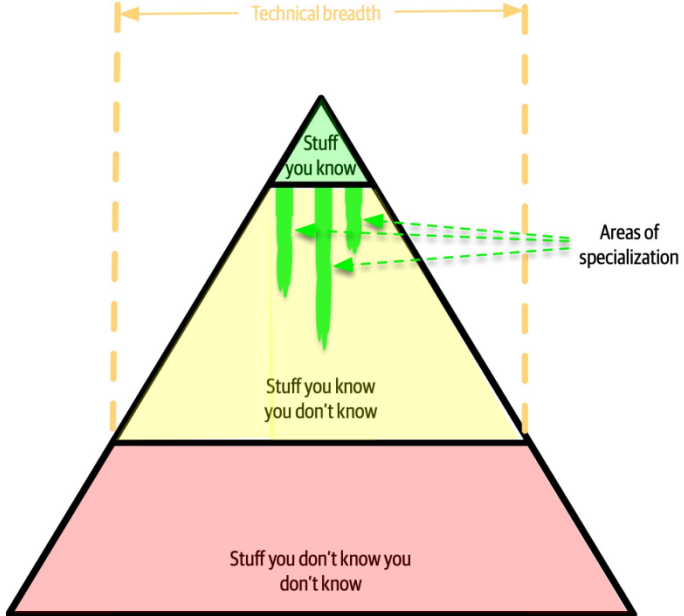
*Architectural Design*

Architects use knowledge and experience to analyse trade-offs to design architectures appropriate to the system context.

# Developers – Technical Depth [Richards and Ford, 2020]



# Architects – Technical Breadth [Richards and Ford, 2020]




- Architects need greater technical breadth than depth
- Breadth allows better consideration of trade-offs
- Avoid trying to become an expert across many areas
  - You'll fail
- Don't stop learning – increase your breadth
  - Don't let your knowledge become stale

*Definition 0.* Conway's Law

Organisations design systems whose structure is inevitably a copy of the organisation's communication structure[Conway, 1968] [MacCormack et al., 2012].

- First citation is original article.
- Second is one of several about MIT and Harvard research into the phenomenon, calling it the *mirroring hypothesis*.
  - Compared open source to commercial packages (e.g. Linux to Solaris).
- Elaborate on this point and Coplien's research into organisational sociology.
- Reasons
  - Governance structures constrain communication paths.
  - This constrains space in which to search for solutions, constraining problem solving approaches.

## *Conway's Law Consequences*

- Business Process Management
- Microservices to reflect organisation structure
- Teams formed around services

- BPM: Redesign organisation structure to reflect system you want
- Microservices: Design system to reflect your organisation
- Elaborate on benefits of both approaches
- Comment on benefits of small focused teams

*Conway's Law Consequences*

## Team insularity – more loyal to team than organisation

- Amazon order tracking example from 2022 week 11
  - Negotiation difficulties with other teams
- Need to ensure inter-team cooperation
- Possibly move people between teams
- Microservices can encourage insularity
- Many teams can also encourage insularity
- Intra-team communication becomes more difficult with large teams

## Conway's Law Issues

- Cross-cutting concerns
  - e.g. Security
- Organisation structure should align with market structure
- Physical location of teams

- Cross-cutting concerns span services & consequently teams
- Can't have a "security" service
  - Needs to be part of *every* service
- Teams solely based around Conway's law and services may not deliver some cross-cutting concerns
- Cooperation, documentation & audits may be necessary
- Market structure may complement team structure to place teams closer to their end users
- Global development & outsourcing mean different teams are likely to be in different locations
- Requires additional overhead & documentation for cooperation between teams

*Definition 0.* Peopleware[Neumann, 1977]

People involved in development of systems.
– Peter G. Neumann

## Stakeholders [Coplien and Bjørnvig, 2010]

- End Users
- Organisation
- Customers / Sponsors
- Domain Experts
- Developers

- Organisation for whom system is being developed
- Customers / Sponsors are those who are responsible for the end user's work process (e.g. managers)
- Domain Experts understand details of problem and/or solution domain

- Written
  - For those who are not there
- Oral
  - Immediate & interactive

- Both apply to all types of stakeholders
- Time Dimension: Written is available in the future
- Extent of either depends on SE process
- Extent also depends on engagement of each type of stakeholder
- Technical Documentation guest lecture in week 11 only touched on docs written for developers
  - Points about clarity are still relevant

## Circumventing Conway's Law [Woods, 2017]

- Cloud Platforms
- Microservices
- APIs
- *Culture*

- Culture is most important
- Easily deployed & accessible services with good APIs opens possibility of communication outside of organisation silos.

*Evidenced-Based Software Engineering*

Don't follow fads, seek evidence for good practice.

Elaborate on finding reliable sources of information and confirming facts yourself.

# *Let's hear from an expert*

## Software Engineering's Greatest Hits

**what we actually know about software development and why we believe it's true**

**Greg Wilson**

http://third-bit.com/talks/greatest-hits/

1 / 47

https://youtu.be/HrVtA-ue-x0

## References


[Conway, 1968] Conway, M. E. (1968).
How do committees invent?
*Datamation*.

[Coplien and Bjørnvig, 2010] Coplien, J. O. and Bjørnvig, G. (2010).
*Learn Architecture for Agile Software Development*.
Wiley.
https://www.amazon.com/Lean-Architecture-Agile-Software-Development/dp/0470684208.

[MacCormack et al., 2012] MacCormack, A., Baldwin, C., and Rusnak, J. (2012).
Exploring the duality between product and organizational architectures: A test of the "mirroring" hypothesis.
*Research Policy*, 41(8):1309–1324.

[Neumann, 1977] Neumann, P. G. (1977).
Peopleware in systems.
*Peopleware in Systems*, pages 15–18.

[Richards and Ford, 2020] Richards, M. and Ford, N. (2020).
*Fundamentals of Software Architecture: An Engineering Approach.*
O'Reilly Media, Inc.

[Woods, 2017] Woods, D. (2017).
How platforms are neutralizing conway's law.
https://www.forbes.com/sites/danwoods/2017/08/15/how-platforms-are-neutralizing-conways-law/.